Closing the gap between perceived and actual waiting times in a call center: results from a field study

Anita Whiting
Department of Marketing, School of Business, Clayton State University, Georgia, USA, and

Naveen Donthu
Department of Marketing, Georgia State University, Atlanta, Georgia, USA

Abstract
Purpose – The purpose of this paper is to investigate what factors influence the gap between caller's perception of how long they think they waited and how long they actually waited on hold and to determine what call managers can do to reduce this gap called estimation error.
Design/methodology/approach – A field experiment was conducted with a corporation's call center.
Findings – The findings were: the higher the estimation error of callers, the less satisfied they are; music increases estimation error, unless callers can choose the music; waiting information reduces estimation error; callers with urgent issues have more estimation error and they overestimate more; and females have higher estimation error and they overestimate more than males.
Research limitations/implications – Limitations are one call center in one context. Implications are identification of antecedents of overestimation.
Practical implications – The paper provides guidelines for call center managers for reducing estimation error and increasing caller satisfaction. It discusses the need for understanding callers and measuring items that are important to them.
Originality/value – The study investigates an under researched variable called estimation error. Study also provides information about some of the causes for why consumers overestimate or underestimate their waiting time. Study provides guidelines from an actual call center and discusses variables that managers can easily use to decrease estimation error and overestimation.

Keywords Call centres, Operating times, Estimation, Individual perception

Paper type Research paper

An executive summary for managers and executive readers can be found at the end of this issue.

Introduction

Call centers have become the dominant form of contact with customers (Micak and Desmarais, 2001). Over 70 percent of customer contact occurs through call centers (Feinberg *et al.*, 2002). Because call centers handle a diverse array of issues ranging from complaint resolution to order taking, call centers have become a critical touch point for managing and increasing customer satisfaction (Anton, 1997; Dawson, 1998). According to Bennington *et al.* (2000, p. 162), call centers have the potential to become the "hub of successful customer relationship management (CRM) strategies and the fulcrum of organizations". Call centers will only continue to grow in importance as more and more companies focus on CRM (Burgers *et al.*, 2000).

With call centers becoming a critical touch point for most organizations, it is important to investigate and understand factors that influence caller satisfaction. Despite the contributions of research on service quality and call centers, there is still a strong need for research in this area (Jack *et al.*, 2006). Organizations with call centers have been criticized for focusing on what is easy to measure (e.g., number of callers served per hour) instead of what is important to measure (e.g., perceived wait time) and for focusing on quantity of calls instead of quality of calls (Robinson and Morley, 2006). Academic literature also lacks knowledge about what is important to caller satisfaction (Feinberg *et al.*, 2002). Most academic studies on call centers have focused on employee issues such as staff dissatisfaction and emotional labor rather than on caller satisfaction (Bennington *et al.*, 2000). Feinberg *et al.* (2002, p. 179) claim that uncovering the significant variables that influence caller satisfaction is "crucial if we are to provide guidance for call center managers". Thus, both managers and academics are very concerned about the lack of knowledge about what influences and drives caller satisfaction.

Within the few studies that have been conducted on caller satisfaction, there is one important variable that has been shown to influence callers and that variable is waiting time. Millions of customers wait on hold in telephone queues to speak to a call center representative (Knott *et al.*, 2004). This

The current issue and full text archive of this journal is available at **www.emeraldinsight.com/0887-6045.htm**

Journal of Services Marketing
23/5 (2009) 279–288
© Emerald Group Publishing Limited [ISSN 0887-6045]
[DOI 10.1108/08876040910973396]

Received: October 2007
Revised: May 2008, September 2008
Accepted: September 2008

Journal of Services Marketing
Volume 23 · Number 5 · 2009 · 279–288

waiting on hold experience has been shown to directly impact satisfaction (Whiting and Donthu, 2006; Antonides *et al.*, 2002; Unzicker, 1999). There are two important variables in a waiting on hold experience. The first variable is the actual (objective) waiting time which is defined as how long the customer actually waited on hold (Hornik, 1984). The second variable is perceived (subjective) waiting time which is defined as how long the customer thinks they waited on hold (Hornik, 1984).

For most consumers, there is usually a gap or discrepancy between actual and perceived waiting time with most consumers overestimating how long they have waited (Hornik, 1984; Katz *et al.*, 1991; Chebat *et al.*, 1991; Knott *et al.*, 2003). This discrepancy between perceived and actual wait times is defined as an estimation error (Knott *et al.*, 2003). Some researchers refer to estimation error as overestimation but some consumers may underestimate their waiting time too. Estimation error is a very important variable because it has been show to influence customer satisfaction (Jones and Peppiatt, 1996).

Because estimation error has been shown to have a significant impact on customer satisfaction, it is important to investigate variables that influence the discrepancy or gap between perceived and actual waiting times within a call center context. Call center managers need to know what variables are causing estimation error and what factors are causing it to increase or decrease. In particular, are there some variables that are causing callers in a call center to overestimate their waiting time while other variables are helping callers to be more accurate in their perceptions of their on hold waiting time? Answering these questions and helping call center managers to decrease estimation error (especially overestimation) is the goal of this research project. In particular, this paper will develop and empirically test a conceptual model that examines determinants of estimation error and its impact on caller satisfaction in a call center. The model contends that real time, expectations, individual differences during the wait, and situational factors during the wait will influence estimation error and satisfaction within a call center context.

This article seeks to make many contributions to the marketing and call center literature. First, this article focuses on perceived wait times, actual wait times, and estimation error within a call center. Most of the research on wait times has focused on either perceived wait times or actual wait times but rarely the discrepancy between the two. Second, this research extends the waiting time literature by investigating the neglected variable called estimation error. Third, this study seeks to explain what factors cause estimation error and why some consumers overestimate their waiting time while others underestimate their waiting times. This research also seeks to add to the literature by investigating waiting times in a new context that is a call center. Most services literature has focused on waiting times in physical settings such as banks, hospitals, and fast food restaurants. However, according to Maister (1985), people will perceive waits differently under different circumstances and therefore, waiting on the telephone may be very different than waiting in an actual service environment. Thus, the findings in a call center may be very different from previous studies in physical service environments. Last, this article provides managerial implications and guidelines to help call center managers decrease estimation error and overestimation.

The article first begins by summarizing the literature on actual waiting times, perceived waiting times, and estimation error. Next, the model is presented and discussed. Third, the article describes the methodology and data collection. Fourth, the article describes the findings from the study and, finally, the article discusses the implications and conclusions from the study and future research opportunities.

Literature review

More and more businesses are adding call centers to their organization. According to the Center for Customer Driven Quality (CCDQ) at Purdue University, the number of call centers has grown from 75,000 in 2001 to an estimated 115,000 in 2005. Approximately 98 percent of *Fortune* 500 companies have call centers (Feinberg *et al.*, 2002). Many organizations are adding call centers because their customers are demanding and expecting telephone access to companies (Cowles and Crosby, 1990).

As the number of call centers continues to grow, businesses must begin to investigate and focus more on managing the on hold telephone wait experience. Waiting on hold to speak to an employee may not be a pleasant experience for some consumers. Many consumers are very conscious of their time costs when waiting (Berry, 1979) and most consumers resent having to wait (Unzicker, 1999). Consumers who have a negative wait experience may even retaliate against businesses by switching to competitors and spreading negative word of mouth (Tom *et al.*, 1997). In order to keep customers happy and satisfied, businesses must be concerned about their customer's waiting on hold telephone experiences.

As discussed previously there are two important variables within an on hold telephone experience. These two variables are actual waiting time and perceived waiting time. Studies on these variables have shown that both influence customer satisfaction (for a review of waiting time literature see Durrande-Moreau (1999)). Estimation error is the difference in perceived and actual wait times.

Estimation error is a very important variable because it occurs very frequently among many consumers. Most estimation error studies have focused on overestimation but consumers may also underestimate their actual waiting time. Research on overestimation error has shown that many consumers greatly overestimate how long they have waited. According to Jones and Peppiatt (1996, p. 47), it is commonly assumed that "the average customer's perception of waiting time is different from reality" with most customers thinking that they have waiting longer than they actually have. Other studies have also found that most consumers overestimate their waiting time. Hornik (1984) conducted a field study on waiting times within the retail industry and found that consumers in a shopping context overestimated their waiting time by 36 percent. Katz *et al.* (1991) found that bank customers overestimated their waiting times by twenty five percent. Jones and Peppiatt (1996) found that their respondents overestimated their wait times by 40 percent. Feinberg and Smith (1989) found that 77 percent of its respondents overestimated their waiting times. Thus, estimation error is occurring in many consumers and the error or discrepancy between actual and perceived waiting time is rather large with most consumer overestimating how long they have waited.

With so many consumers experiencing estimation error and by such a large percentage, it is important to investigate what drives the discrepancy between perceived and actual waiting times especially within a call center context. As previously discussed there are only a few studies that have investigated estimation error. However, these studies did not investigate the causes of estimation error and they did not investigate estimation error within a call center context. Most waiting time studies have focused on the customer's perception of the wait (and not actual wait time) and most studies have focused on waiting within a service setting (e.g., bank or store) and not on the telephone. Most wait studies collected perceived wait times but they did not measure actual wait times; and thus did not investigate estimation error (Jones and Peppiatt, 1996). The lack of literature on estimation error may be due to the challenges of collecting actual wait times from consumers. This article seeks to address this gap in the literature by developing a model of determinants that influence estimation error and caller satisfaction within a call center context.

Model development

In order to investigate the determinants of estimation error, we chose to rely on Durrande-Moreau's (1999) review of the waiting literature. She reviewed over 30 papers on wait management between the years of 1984 through 1997 and she concluded that there are six factors that influence consumers while waiting. These six factors are:

1 real time;
2 personal expectations;
3 individual factors before the wait;
4 situational factors before the wait;
5 individual factors during the wait; and
6 situational factors during the wait.

Because factors before the wait cannot be easily controlled by call center managers, we chose to focus on four variables that can be managed and their impact on estimation error and caller satisfaction. The four variables are:

1 real time;
2 personal expectations;
3 individual factors during the wait; and
4 situational factors during the wait.

The effects of these four variables will be explained by applying Zakay's (1989) Resource Allocation model (see Figure 1).

It is important to note that Durrande-Moreau's review did not find estimation error to be a frequently investigated variable. Most of the studies reviewed by Durrande-Moreau were focused on perceived wait times and satisfaction. This study builds upon and extends Durrande-Moreau's review by investigating four of her six variables and their impact on estimation error.

Real time

Real time has been shown to have a negative impact on the waiting experience. According to Durrande-Moreau's (1999) review, real time was the central stimulus for reactions to the wait and that the longer the duration, the more negative the reaction to the wait. In addition to satisfaction, real time may also have an impact on estimation error. Studies on estimation error have found that shorter the wait, the greater the estimation error (Davis and Volman, 1990; Jones and Peppiatt, 1996). Evangelist *et al.* (2002) found that customers with waits of less than three minutes were more likely to overestimate their waiting time while customers with waits greater than five minutes were more likely to underestimate their waiting time.

The inverse effect of real time on estimation error can be explained by Zakay's (1989) Resource Allocation model. Zakay's model proposes that time perception is a function of the number of time units recorded by a cognitive timer. This cognitive timer is activated when people pay attention to the passage of time. At the beginning of the waiting experience, consumers are occupied with the passage of time and they actively engage in time estimations. However, as the wait continues, consumers become distracted by stimuli and they begin to make fewer time estimations. These fewer wait estimations lead to more accurate perceptions of the wait time or even under evaluations of the wait time. Based on Zakay's model and on the findings in physical service settings, the following hypothesis is proposed:

H1. The longer the actual waiting time in a call center, the lower the estimation error.

Estimation error has also been shown to influence consumer satisfaction. Jones and Peppiatt (1996) investigated the gap between actual and perceived waiting times and found that estimation error had an impact on satisfaction. In particular, they found that higher estimation error leads to less satisfaction. This inverse relationship between estimation error and satisfaction can be explained by Parasuraman *et al.*'s (1985) widely accepted service quality model. According to the model, there is a gap between actual delivery of service (actual wait time) and customer's perception of the service (perceived wait time). This gap along with the other gaps has a negative influence on customer satisfaction and service quality. Therefore, the following hypothesis is proposed:

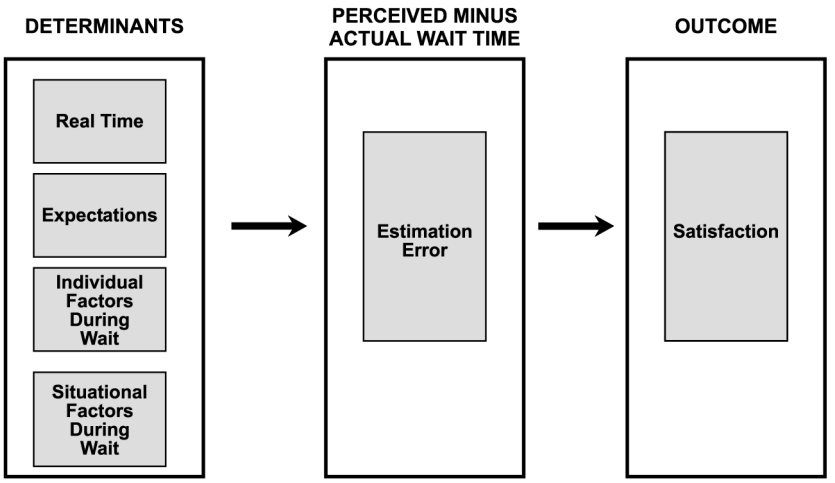
H2. The higher the estimation error of a caller, the lower the caller's satisfaction.

Personal expectations

According to Durrande-Moreau's (1999) review, personal expectations strongly influence outcome variables. In her review of 18 articles on expectations and waiting time, she observed a "classical comparative mechanism between expectation and reality that exemplifies the confirmation-disconfirmation paradigm" (Durrande-Moreau, 1999, p. 175). She also reported that customers who expect a short wait will react more negatively than others. We predict that expectations of a short wait will have a negative impact on estimation error and satisfaction. This prediction is based on the discrepancy theory (Michalos, 1985) and the expectancy disconfirmation paradigm. These theories suggest that consumers establish expectations, observe the performance, compare the performance to expectations, and then form disconfirmation perceptions (Yan and Lotz, 2006). When the disconfirmation between expectations of the wait and the actual wait time is large, the estimation error will be large and the consumer will be less satisfied. The following hypotheses are therefore proposed:

H3. Individuals with expectations of a short wait will have higher estimation error than those with expectations of a longer wait.

Figure 1 Model of estimation error in a call center



H4. Individuals with expectations of a short wait will have lower customer satisfaction scores than those with expectations of a longer wait.

Individual factors during the wait

Maister (1985) proposed that people will perceive waits differently. There have been many individual factors reported to influence waiting times such as type of customers (experienced vs novice), value of purchase, and time pressure. Based on Durrande-Moreau's (1999) review of individual factors, we chose to look at music preference, gender, and experience and their impact on estimation error and satisfaction.

Music played during the wait has been shown to influence waiting times. In particular, music has been shown to influence perceived waiting duration and behavior (Hui *et al.*, 1997). Music adds to the service environment and helps create a more positive evaluation (Baker *et al.*, 1992). Both Kellaris and Kent (1992) and Katz *et al.* (1991) found that playing music reduces the negative effects of waiting. North *et al.* (1999) investigated the effects of liking and fit of music on the amount of time callers would stay on hold. He found that callers would wait on hold longer when music they liked was played. The beneficial effects of liked music can be explained by Zakay's (1989) Resource Allocation model. According to the model, consumers are occupied with the passage of time and they actively engage in time estimations. However, when liked music is played, consumers become distracted by the music and they begin to make fewer time estimations. These fewer wait estimations lead to more accurate perceptions of the wait time or even under evaluations of the wait time. We therefore predict that liked music will have a positive influence on estimation error. The following hypothesis is proposed:

H5. Callers who like the music played will have lower estimation error than callers who don't like the music played.

Gender is also another individual factor that may influence estimation error. Gender has been shown to influence many outcome variables. According to Karatepe *et al.* (2006, p. 1088) there is a distinction between how each gender "observes the environment, processes, evaluates and retrieves information, and makes judgments". Women look at the details and process lots of information when making decisions while men use heuristics and process less information (Sunden and Surette, 1998). It has also been shown that females experience higher levels of stress (Nelson and Quick, 1985). Based on these findings, we predict that gender will influence estimation error. The following hypothesis is proposed:

H6. Females will have higher estimation error than males.

Despite the previous findings on gender, we acknowledge that other studies have found contradictory results showing that there are no differences between men and women and their waiting experiences. Both Davis and Volman (1990) and Jones and Peppiatt (1996) did not find any gender differences in their waiting studies. However, there were many other variables investigated in their studies which may have caused noise in the data and thus caused the gender differences not to come through.

Experience may also play a role in estimation error. Customer's prior experience has been shown to influence both perceived wait and satisfaction (Davis and Volman, 1990). Jones and Peppiatt (1996) found that new or infrequent users had significantly higher perceived wait times than frequent users. Customers' prior experience can also by explained by Zakay's Resource Allocation model. New and inexperienced users may focus on the passage of time and actively engage in time estimations. Experienced users may not engage in as many time estimations because they are familiar with the waiting situation. We therefore predict that lack of experience will have a negative impact on estimation error. The following hypothesis is proposed:

H7. Novice callers will have higher estimation error than experienced callers.

Situational factors during the wait

Durrande-Moreau's (1999) found that situational factors were the most examined factor and that many of them have an influence on consumers. Some of the situational variables investigated have been type of queue, television, and information displays. For this study, we chose to focus on

presence of music, waiting information given, and urgency of the call.

The presence of music has been shown to impact many important dependent variables such as length of time in store, amount purchased, and likelihood of returning (Oakes, 2000). Presence of music differs from the previously mentioned variable about liking the music that is played. Liking the music played is an individual factor while presence of music is a situational factor that deals only with the presence or absence of music. Research on the presence of background music has been shown that it has positive effects on consumers by decreasing stress and increasing relaxation (Tansik and Routhieaux, 1999). Research on music has also shown that there is a significant relationship between waiting and music (Chebat *et al.*, 1993). Music has been shown to reduce perceived wait times in restaurants and supermarkets (Milliman, 1982, 1986). These positive effects can be explained by Zakay's Resource Allocation model. The presence of music may distract individuals from the passage of time and it may cause them not to engage in as many time estimations. These fewer time estimations may lead to more accurate perceptions of the wait time or even under evaluations of the wait time. We therefore predict that the presence of music will decrease estimation error. The following hypothesis is proposed:

H8. Callers with background music will have lower estimation error than callers without background music.

In addition to music, waiting information has also been shown to influence consumer's perception of the wait (Ahmadi, 1984; Katz *et al.*, 1991). There are two types of waiting information: estimated wait time and number in the queue. Estimated wait time is information about the expected length of the wait and queuing information is the consumer's position in the queue (Hui and Tse, 1996). According to Maister (1985) uncertain waits are longer than known waits. Zakay and Hornik (1994) suggest that information about the wait reduces consumers from thinking about how long they have been waiting and thus reduces their perception of the waiting time. Based on these findings, we predict that waiting information will have a positive impact on estimation error. The following hypothesis is proposed:

H9. Callers with waiting information will have lower estimation error than callers without waiting information.

Urgency of the call may influence estimation error. Criticality of time to the customer has been shown to influence perception of the wait and satisfaction (Davis and Volman, 1990). Davis and Heineke (1998) found that satisfaction with the wait depends on the differences in the needs of the consumer. According to Maister (1985), people perceive waits differently under different situations such as an urgent situation. Maister also proposes that uncomfortable waits (such as an urgent call) seem longer than comfortable waits. The relationship between urgency and estimation error can be explained by Zakay's Resource Allocation model. Individuals with urgent issues are very focused on the passage of time and they are constantly making time estimations. These frequent time estimations may lead to very inaccurate accurate perceptions of the wait time with most urgent callers greatly overestimating their wait time. We therefore predict that urgency of the call will have a negative impact on estimation error. The following hypothesis is proposed:

H10. Callers with urgent issues will have higher estimation error than callers with nonurgent issues.

Methodology

Overview

A national corporation agreed to let us use their call center to collect data. Their call center supports franchisees with their point of sales systems and their back office systems. The respondents in this study were independent franchise owners who paid monthly fees for the services provided by the call center. The call center was currently using background music and two information cues and they wanted to see how these variables were affecting their callers. Therefore, we conducted an experiment to investigate these situational factors while also gathering data on other variables.

For the experiment, we manipulated music, estimated wait time given, and number in the queue given. The experiment consisted of eight different treatments. The treatments were:

1 no music, no information;
2 with music, no information;
3 no music, estimated wait time given;
4 with music, estimated wait time given;
5 no music, number in queue given;
6 with music, number in the queue given;
7 no music, both estimated wait time and number in queue given; and
8 with music, both estimated wait time and number in queue given.

The experiment was conducted over an eight-week period with each week being a different treatment.

Procedure

Franchisees would call into the call center for assistance with issues about their point of sales systems or their back office systems. While they were on hold, the callers were exposed to the treatment for that week. The experiment was conducted over eight weeks and measures were taken to ensure that survey respondents were only questioned once about their waiting hold experience. Call center employees were instructed to write down the actual waiting time of each caller and the store's number (both of which were on the computer screen) as they answered each call. The employees also wrote down the caller's name. Later that evening, an e-mail survey was sent to the franchisee at the store's e-mail address. The e-mails were sent out from a university e-mail address so that callers could be more candid with their responses. Reminder e-mails were also sent out a day after the initial e-mail. A total of 211 completed e-mail surveys were returned. The response rate was approximately 18 percent.

Measures

The survey consisted of 14 questions. Participants were asked to state the reason for their call and they were asked how long they think they waited on the phone before an agent answered the call. Participants were also surveyed about the presence of music and their feelings about the music. Additional questions on the survey were about expectations about the wait and satisfaction with the wait. Actual wait time data and perceived wait time data were matched up for each caller and

thus estimation error was calculated. The company also provided additional information to help with the data analysis. Gender and years of experience as franchisee (based on company records) were provided by the organization. A supervisor in the call center categorized the data from the reason for the call question into two categories: urgent and nonurgent calls.

Data analysis and results

We first began our data analysis by analyzing the average wait times for each treatment group. We did not find actual wait times for each treatment group to be significantly different; so, we were able to move forward with analyzing the data. For each treatment, we calculated estimation error by subtracting the actual wait times from the perceived wait times. We conducted an ANOVA on estimation error and found that estimation error approached significance at the 0.08 level among all eight treatments. There were also significant differences among some of the treatments but this will be discussed later under situational factors. We also conducted an ANOVA on perceived waiting times and found that there were no significant differences in perceived wait times among the treatments. Thus estimation error was impacted more than perceived wait times by the experimental variables of music and information cues.

The average wait time for the entire sample was 1 minute and 38 seconds with actual wait times ranging from 12 seconds to 8 minutes. The average perceived wait time for the entire sample was 3 minutes and 23 seconds with perceived wait times ranging from 20 seconds to 15 minutes. The average estimation error for the entire sample was 1 minute and 39 seconds with estimation error ranging from less than 4 minutes to 10 minutes. Approximately 79 percent of the sample overestimated their waiting time. The average estimation error for each of the eight treatment groups is shown in Table I.

Next we analyzed the data from our hypotheses on real time, personal expectations, individual factors during the wait, and situational factors during the wait. Our empirical findings of these four variables on estimation error are discussed below (see Table II).

Real time

We first investigated the actual length of the wait and its impact on estimation error and found a significant negative correlation between the two (−0.243 correlation which is significant at 0.05 level). Thus, *H1* was supported and the finding was that as the actual wait increases, estimation error declines. Next, we investigated the relationship between estimation error and caller satisfaction.

As predicted by *H2*, estimation error had a significant negative correlation with satisfaction (−0.280 which is significant at 0.01 level). Thus, the more estimation error, the less satisfied they are.

Table I Average estimation error in experimental treatments

Treatment	Average estimation error
(1) No music, no information	2.25
(2) With music, no information	2.34
(3) No music, estimated wait time given	−0.34
(4) With music, estimated wait time given	2.07
(5) No music, number in queue given	0.36
(6) With music, number in the queue given	2.59
(7) No music, both estimated wait time and number in queue given	0.37
(8) With music, both estimated wait time and number in queue given	1.70

Expectations

Next, we investigated how expectations about the wait would impact estimation error and satisfaction. Findings from an ANOVA tests showed that there we no significant differences in estimation error among those whose waits were more than expected, about what expected, and less than expected. Thus, *H3* was not supported. We also investigated how expectations impact satisfaction using ANOVA. We found there to be a significant difference in satisfaction scores among consumers with difference expectations (0.001 sign). Thus, *H4* was supported. Overall, we found expectations about the wait did not impact estimation error but it did greatly impact satisfaction.

Individual differences during the wait

The three individual differences that we investigated were feelings about the music, gender, and experience (novice vs experienced user). An ANOVA test on feelings about music and estimation error was significant at 0.027. The estimation error scores were 1.88 for no music, 0.275 for liked music, and 2.18 for does not like music. Pair wise comparison tests showed a 0.021 significant difference between no music and liked music and a 0.011 significant difference between liked music and does not like music. Thus, *H5* was supported and the major finding was that estimation error is significantly less for consumers who like the music that is played. For gender, we conducted another ANOVA on estimation error. From this analysis, we found gender to be significant at 0.016. The average estimation error score was 0.559 for males was and 2.27 for females. Thus *H6* was supported and the major finding was that females have higher estimation error scores than males and they significantly overestimate wait times more than males. For experience, we categorized our respondents into two categories: novice (0-2 years of experience) and experienced (three to 30 years). Despite correlation and ANOVA tests, we did not find experience to influence estimation error or satisfaction. Thus, *H7* was not supported.

Situational factors during the wait

The three situational variables that we investigated were:

1 presence of music;
2 waiting information given; and
3 urgency of call.

Presence of music was manipulated in the experiment with callers with four of the eight treatments having music played in the background while waiting on hold. An ANOVA was conducted on estimation error with treatments as the independent variable. Treatments approached significance at the 0.08 level. Pair wise comparisons of the different treatment groups yielded significant findings. Treatment 3 (no music and estimated wait time given) and treatment 4 (with music and estimated wait time given) were significantly

Table II Summary of hypotheses and findings

Hypotheses	Findings
Real time	
1 The longer the actual waiting time in a call center, the less the estimation error	Supported at .05 level
2 The higher the estimation error of a caller, the lower the caller's satisfaction	
Expectations	
3 Individuals with expectations of a short wait will have higher estimation error than those with expectations of a longer wait	Not supported
4 Individuals with expectations of a short wait will have lower customer satisfaction than those with expectations of a longer wait	Supported at 0.01 level
Individual factors	
5 Callers who like the music played will have lower estimation error than callers who do not like the music played	Supported at 0.05 level
6 Females will have higher estimation error than male	Supported at 0.05 level
7 Novice callers will have higher estimation error than experienced callers	Not supported
Situational factors	
8 Callers with background music will have lower estimation error than callers without background music	Supported at 0.05 level
9 Callers with waiting information will have lower estimation error than callers without waiting information	Supported at 0.05 level
10 Callers with urgent issues will have higher estimation error than callers with non urgent issues	Supported at 0.05 level

different at the 0.05 level. The estimation error values for treatments 3 and 4 are −0.34 and 2.07 respectively. When treatment 5 (no music and number in queue information given) was compared to treatment 6 (with music and number in queue information given), a 0.26 significant difference was found. The estimation error values for treatment 5 and 6 are 0.36 and 2.59 respectively. These findings demonstrate that the presence of music actually increases estimation error. Thus, *H8* was not supported. The overall finding from this analysis is that the presence of music greatly increases estimation error among callers.

Waiting information

It was also manipulated in the experiment. The four categories were:

1 no information given;
2 estimated wait time given;
3 number in queue information given; and
4 both estimated wait time and number in queue information given.

An ANOVA was conducted on overestimation with treatments being the independent variable. Treatments approached significance at the 0.08 level. Pair wise comparisons of the different treatment groups against the control group yielded three significant findings. Treatment 1 (no music and no information) and treatment 3 (no music and wait information given) were significantly different at the 0.003 level. The estimation error values were 2.25 and −0.34 respectively. Thus, estimated wait time information significantly reduced estimation error among callers. The second pair wise analysis showed significant differences at the 0.014 level between treatment 1 (no music and no information) and treatment 5 (no music and number in queue information given). The estimation error values for treatments 1 and 5 are 2.25 and 0.36 respectively. Thus, the second finding is that number in queue information greatly reduces estimation error among callers. The third pair wise analysis showed significant differences at the 0.046 level among treatment 1(no music and no information) and treatment 7 (no music and both estimated wait time and number in queue given). The estimation error values for treatments 1 and 7 are 2.25 and 0.37. Thus, both estimated wait time and number in queue information greatly reduced estimation error among callers. Thus, *H9* was supported and the overall finding from these three analyses was that waiting information does in fact greatly reduce estimation error among callers.

Urgency of call

An ANOVA was conducted on estimation error with urgency of call as the independent variable. Urgency of the call was found to be significant at the 0.013 level. The average estimation error for nonurgent calls was 1.11 while average estimation error for urgent calls was 2.60. Thus, *H10* was supported and the overall finding was that callers with more urgent issues are more likely to have more estimation error and are also more likely to inflate how long they have waited on hold.

Discussion and implications

An experiment was conducted at a call center to understand what drives the gap between perceived and actual waiting times in a call center. This was done in order to help call center managers understand the huge impact that estimation error has on caller satisfaction and to provide guidelines for reducing estimation error and increasing satisfaction. Call centers are a critical customer touch point that must be managed. Managers of call centers have to not only focus on reducing the actual waiting time for callers put on hold, but they must also manage and align the perception of wait time more closely with the actual waiting time, and thus decrease estimation error. Managers must focus on closing the gap between perception and reality.

From our study, we found eight major findings. These findings are very actionable and they have major implications for call center managers. The major findings are:

- The larger the gap between perceived and actual waiting times, the less satisfied consumers are.

- As actual wait increases, estimation error declines.
- Expectations about the wait do not impact estimation error, but it does greatly impact customer satisfaction.
- Presence of music greatly increases estimation error among callers.
- However, estimation error of waiting times is significantly less for consumers who like the music that is played.
- Compared to males, females have significantly more estimation error and they overestimate their wait time more.
- Waiting information does greatly reduce estimation error and overestimation among callers.
- Callers with more urgent issues have more estimation error are more likely to inflate how long they have waited on hold.

Implications

First, call centers are a major customer touch point that must be managed and it must be managed well. A majority of customer contact occurs through them and the image of the company from the customer's eyes can either be demolished or enhanced from interactions with a call center. There are major consequences for companies that do not focus on caller satisfaction. Dissatisfied customers are more likely to spend less, switch to competitors, and spread negative word of mouth communication. Organizations that do not focus heavily on caller satisfaction may want to evaluate how much and how important the business activities are that occur through their call centers. For organizations that conduct critical and/or numerous activities through their call center, the stakes are much higher for increasing caller satisfaction.

Second, organizations must focus on estimation error. Estimation error occurs frequently with most consumers thinking that they have waited significantly longer than they actually have. Decreasing estimation error and overestimation should be an initiative for organizations because of its direct impact on customer satisfaction. Organizations must also collect data on perception of the wait and estimation error (and not just collect actual wait time data because of its convenience). Information on perceptions of the wait and estimation error is important because consumers may feel like a one-minute wait lasted ten minutes. Organizations may also want to rethink the data (e.g., number of calls handled per employee) that they are collecting to see if that data are really important to customers. From the study we found that actual wait times did not influence customer satisfaction, but perception of the wait and most significantly estimation error of the wait did. However, not many organizations are currently collecting this data.

Third, just playing music while callers are put on hold is not adequate. Music should be liked by the callers for it to reduce the estimation error of waiting time. Organizations should research their customers and match the music to the customer's preferences. For organizations with many different types of consumers, they may want to use a system that would give callers a choice of music. Organizations may also want to consider the option of silence or the news. Silence may be soothing for stressed customers and it may be better than listening to something that is not liked. News may also be a good option for consumers who want to be informed and educated while waiting on hold instead of just wasting time.

Fourth, organizations should consider providing information cues such as estimated wait times. Information cues reduce estimation error and it helps callers be more realistic about how long they actually waited. Providing waiting information also helps reduce uncertainty and it helps consumers decide whether or not to wait on hold. Consumers who hear an estimated wait time of 25 minutes may quickly decide to call back later. Estimated wait times also set consumers' expectations for how long they will have to wait. With telephone holds, consumers can not see the virtual line of callers in front of them and they have no way of knowing how long the wait may be. Managing customer's expectations is critical for providing good customer service.

Fifth, urgent callers should be given special attention and if possible not be put on hold. One way to address this is by letting callers specify why they are calling via a menu selection. An example of this would be press 1 for computer problems, 2 for payroll problems, and 3 for e-mail problems. More urgent calls should be answered first. Organizations may also use different telephone numbers for different types of calls.

Sixth, organizations need to understand their callers. This study and a study by Jones and Peppiatt (1996) found that females had more estimation error and they overestimated more than males. Thus, organizations need to know the gender of their target audience. Retailers whose customer base is mostly women need to focus heavily on perception techniques. The waiting strategies of organizations must line up with the target market that it serves.

Overall, the theme that emerges from these implications is that organizations must understand who the caller is. Organizations need to know their gender, music preference, and urgency of the call. By using customer profiling, the call center can do every thing possible to reduce estimation error and overestimation of waiting times. Reducing estimation error and overestimation of waiting times is key to increasing caller satisfaction. From CRM initiatives, organizations are moving toward understanding and creating relationships with its customers. Call centers are a crucial touch point where relationships can be created and grown.

Limitations and future research

Even though this study was conducted in a corporate call center, it still has some limitations. First, the current study findings are based on one experiment that was conducted in one call center. Different types of call centers may produce different results. Another limitation is that waits of different lengths may produce different responses. The average wait time in this study was approximately two minutes. Future work should increase the generalizability of the findings by testing in other contexts with different samples and with different waiting times. Despite these limitations, this field study demonstrated that organizations can close the gap between perception and reality and decrease overestimation.

Because estimation error occurs with almost all consumers, it is an area that must be further investigated. Businesses need to understand the causes for people inflating their waiting time. This study only looked at a few causes of estimation error but there are lots more to investigate. In particular, future studies could look at additional individual and situational factors. For individual factors, researchers could look at different types of callers, frequency of the caller,

reason for call, and age. Additional individual factors to explore are callers that balk and do not even wait on hold and callers that renege and hang up before their call is answered. There is much to be gained by understanding why and which type of callers hangs up before their call is answered. There are also many situational factors to be investigated. Music has many different components (e.g., tempo, volume, type, etc.) that could be researched. Researchers could also explore reason for call (e.g., billing, making purchase, complaining), day of week, and time of day. There are also additional audio items to be investigated such as announcements or advertisements to callers while holding. Caller satisfaction and estimation error is an under researched area that has endless opportunities. The importance of this research area will only continue to grow as more and more organizations focus on CRM and try to increase customer satisfaction.

References

Ahmadi, K. (1984), "Effects of social influences and waiting on time judgment", *Perceptual and Motor Skills*, Vol. 59, pp. 771-6.

Anton, J. (1997), *Call Center Management by the Numbers*, Purdue University Press, West Lafayette, IN.

Antonides, G., Verhoef, P.C. and Aalst, M. (2002), "Consumer perception and evaluation of waiting time: a field experiment", *Journal of Consumer Psychology*, Vol. 12 No. 3, pp. 193-202.

Baker, J., Levy, M. and Grewal, D. (1992), "An experimental approach to making retail store environment decisions", *Journal of Retailing*, Vol. 68 No. 4, pp. 445-60.

Bennington, L., Cummane, J. and Conn, P. (2000), "Customer satisfaction and call centers: an Australian study", *International Journal of Service Industry Management*, Vol. 11 No. 2, pp. 162-73.

Berry, L.L. (1979), "The time buying consumer", *Journal of Retailing*, Vol. 55 No. 4, pp. 58-69.

Burgers, A., Ruyter, K., Keen, C. and Streukens, S. (2000), "Customer expectation dimensions of voice to voice service encounters: a scale development study", *International Journal of Service Industry Management*, Vol. 11 No. 2, pp. 142-61.

Chebat, J.C., Filiatrault, P. and Zuccaro, C. (1991), "An experiment on waiting lines: effects of interrupted service and clients' participation on perceived time duration, mood, and perceived quality", in Chebat, J.C. and Venkatesan, V. (Eds), *Proceedings of the VIIth John-Labatt Marketing Research Seminar, Time and Consumer Behavior, UQAM, Montreal, Canada*.

Chebat, J.C., Gelinas-Chebat, C. and Filitrault, P. (1993), "Interactive effects of music and visual cues on time perception: an application to waiting lines in banks", *Perceptual and Motor Skills*, Vol. 77, pp. 995-1020.

Cowles, D. and Crosby, L.A. (1990), "Consumer acceptance of interactive media", *The Services Industries Journal*, Vol. 10 No. 3, pp. 521-40.

Davis, M. and Heineke, J. (1998), "How disconfirmation, perception, and actual waiting times impact customer satisfaction", *International Journal of Service Industry Management*, Vol. 9 No. 1, pp. 64-73.

Davis, M. and Volman, T. (1990), "A framework for relating waiting time and customer satisfaction in a service operation", *Journal of Service Marketing*, Vol. 4 No. 1, pp. 61-9.

Dawson, K. (1998), *The Call Center Handbook: The Complete Guide to Starting, Running, and Improving Your Call Center*, Miller-Freeman, New York, NY.

Durrande-Moreau, A. (1999), "Waiting for service: ten years of empirical research", *International Journal of Service Industry Management*, Vol. 10 No. 2, pp. 171-89.

Evangelist, S., Godwin, B., Johnson, J., Conzola, V., Kizer, R., Young-Helou, S. and Metters, R. (2002), "Linking marketing and operations: an application at Blockbusters, Inc.", *Journal of Service Research*, Vol. 5 No. 2, pp. 91-100.

Feinberg, R.A. and Smith, P. (1989), "Misperception of time in the sales transaction", in Srull, T.K. (Ed.), *Advances in Consumer Research*, Vol. 16, Association of Consumer Research, Provo, UT, pp. 56-8.

Feinberg, R., Hokama, L., Kadam, R. and Kim, I. (2002), "Operational determinants of caller satisfaction in the banking/financial service call center", *International Journal of Bank Marketing*, Vol. 20 No. 4, pp. 174-80.

Hornik, J. (1984), "Subjective vs objective time measures: a note on the perception of time in consumer behavior", *Journal of Consumer Research*, Vol. 11 No. 1, pp. 615-8.

Hui, M. and Tse, D. (1996), "What to tell consumers in waits of different lengths: an integrative model of service evaluation", *Journal of Marketing*, Vol. 60 No. 2, pp. 81-90.

Hui, M.K., Dube, L. and Chebat, J.C. (1997), "The impact of music on consumer's reactions to waiting for service", *Journal of Retailing*, Vol. 73 No. 1, pp. 87-104.

Jack, E.P., Bedics, T.A. and McCary, C.E. (2006), "Operational challenges in the call center industry: a case study and resource based framework", *Managing Service Quality*, Vol. 16 No. 5, pp. 477-500.

Jones, P. and Peppiatt, E. (1996), "Managing perceptions of waiting times in service queues", *International Journal of Service Industry Management*, Vol. 7 No. 5, pp. 47-61.

Karatepe, O.M., Yavas, U., Babakus, E. and Avci, T. (2006), "Does gender moderate the effects of role stress in frontline service jobs?", *Journal of Business Research*, Vol. 59 Nos 10/11, pp. 1087-93.

Katz, K., Larson, B. and Larson, R. (1991), "Prescription for the waiting-in-line blues: entertain, enlighten and engage", *Sloan Management Review*, Vol. 32 No. 2, pp. 44-53.

Kellaris, J. and Kent, R. (1992), "The influence of music in consumers' temporal perceptions: does time fly when you're having fun", *Journal of Consumer Psychology*, Vol. 4 No. 1, pp. 365-76.

Knott, B.A., Kortum, P., Bushey, R.R. and Bias, R. (2004), "The effect of music choice and announcement duration on subjective wait time for call center hold queues", *Proceedings of the Human Factors and Ergonomics Society 47th Annual Meeting*.

Knott, B.A., Pasquale, T., Miller, J., Mills, S. and Joseph, K.M. (2003), "Please hold for the next available agent: the effect of hold queue content on apparent hold duration", *Proceedings of the Human Factors and Ergonomics Society 47th Annual Meeting, Denver, CO.*

Maister, D. (1985), "The psychology of waiting lines", in Czepiel, J., Solomon, M. and Suprenant, C. (Eds), *The Service Encounter*, Lexington Books, Lexington, MA.

Micak, A. and Desmarais, M. (2001), "Benchmarking service quality performance at business-to-business and business-to-consumer call centers", *Journal of Business & Industrial Marketing*, Vol. 16 No. 5, pp. 340-53.

Michalos, A.C. (1985), "Multiple Discrepancies Theory (MDT)", *Social Indicators Research*, Vol. 16, pp. 347-413.

Milliman, R.E. (1986), "The influence of background music on the behavior of restaurant patrons", *Journal of Consumer Research*, Vol. 13, September, pp. 286-9.

Milliman, R.E. (1982), "Using background music to affect the behavior of supermarket shoppers", *Journal of Marketing*, Vol. 46 No. 2, pp. 86-91.

Nelson, D.L. and Quick, J.C. (1985), "Professional women: are distress and disease inevitable", *Academy Management Review*, Vol. 10 No. 2, pp. 206-18.

North, A., Hargreaves, D. and McKendrick, J. (1999), "Music and on-hold waiting time", *British Journal of Psychological*, Vol. 90 No. 1, pp. 161-4.

Oakes, S. (2000), "The influence of the musicscape within service environments", *Journal of Service Marketing*, Vol. 14 No. 7, pp. 539-56.

Parasuraman, A., Zeithaml, V. and Berry, L. (1985), "A conceptual model of service quality and its implications for future research", *Journal of Marketing*, Vol. 49 No. 4, pp. 41-50.

Robinson, G. and Morley, C. (2006), "Call centre management: responsibilities and performance", *International Journal of Service Industry Management*, Vol. 17 No. 3, pp. 284-300.

Sunden, A. and Surette, B. (1998), "Gender differences in the allocation of assets in retirement savings plans", *American Economic Review*, Vol. 88 No. 2, pp. 207-12.

Tansik, D. and Routhieaux, R. (1999), "Customer stress relaxation: the impact of music in a hospital waiting room", *International Journal of Service Industry Management*, Vol. 10 No. 1, pp. 68-81.

Tom, G., Burns, M. and Zeng, T. (1997), "Your life on hold: the effect of telephone waiting time on consumer perception", *Journal of Direct Marketing*, Vol. 11 No. 3, pp. 25-31.

Unzicker, D. (1999), "The psychology of being put on hold: an exploratory study of service quality", *Psychology and Marketing*, Vol. 16 No. 4, pp. 327-50.

Whiting, A. and Donthu, N. (2006), "Managing voice-to-voice encounters: reducing the agony of being put on hold", *Journal of Service Research*, Vol. 8 No. 3, pp. 234-44.

Yan, R. and Lotz, S. (2006), "The waiting game: the role of predicted value, wait disconfirmation, and providers' actions in consumers' service evaluations", in Kardes, F. and Mita, S. (Eds), *Advances in Consumer Research*, Vol. 22, Association for Consumer Research, Provo, UT, pp. 412-8.

Zakay, D. (1989), "Subjective time and attentional resource allocation: an integrated model of time and estimation", in Levin, I. and Zakay, D. (Eds), *Time and Human Cognition*, North Holland, Amsterdam, pp. 365-98.

Zakay, D. and Hornik, J. (1994), "How much time did you wait in line? A time perception perspective", paper presented at the 6th John-Labatt Marketing Research Seminar, University of Quebec in Montreal, Montreal.

About the authors

Anita Whiting is an Assistant Professor of Marketing at Clayton State University. She received her PhD in Marketing from Georgia State University. Dr Whiting also holds an MBA from the Georgia Institute of Technology. She has published in *Journal of Service Research*, *Journal of Services Marketing*, and *International Business: Research, Teaching, and Practice*. Services marketing, retailing, frontline service employees, and atmospherics are her major research interests. Anita Whiting is the corresponding author and can be contacted at: AWhiting@clayton.edu

Naveen Donthu (PhD University of Texas at Austin) is the Katherine S. Bernhardt Research Professor of Marketing at Georgia State University. His expertise is in the areas of marketing research methodology, consumer research, advertising, electronic business and services marketing. He has published over 65 articles in peer reviewed journals, including *Journal of Marketing*, *Journal of Marketing Research*, *Journal of Consumer Research*, *Marketing Science* and *Management Science*.

To purchase reprints of this article please e-mail: **reprints@emeraldinsight.com**
Or visit our web site for further details: **www.emeraldinsight.com/reprints**

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.